EXPERIMENT : 10

Performance Evaluation of NOMA and Successive Interference Cancellation.

OBJECTIVES :

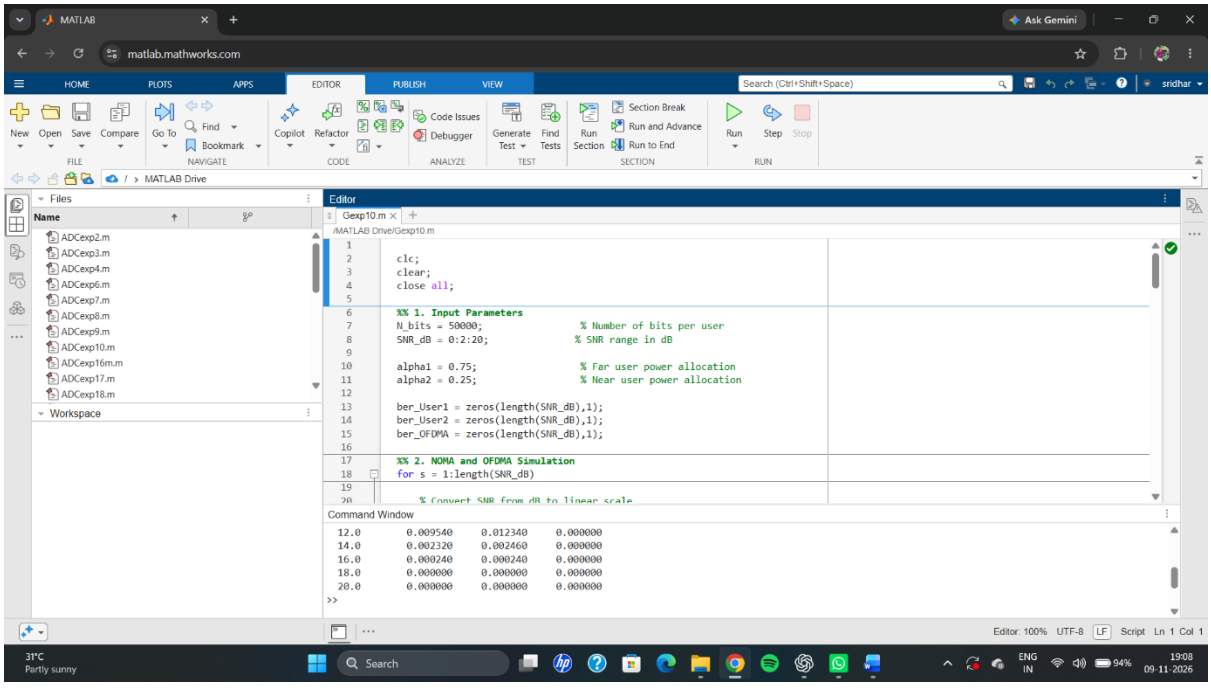
1. To model a two-user power-domain NOMA transmission system using MATLAB.

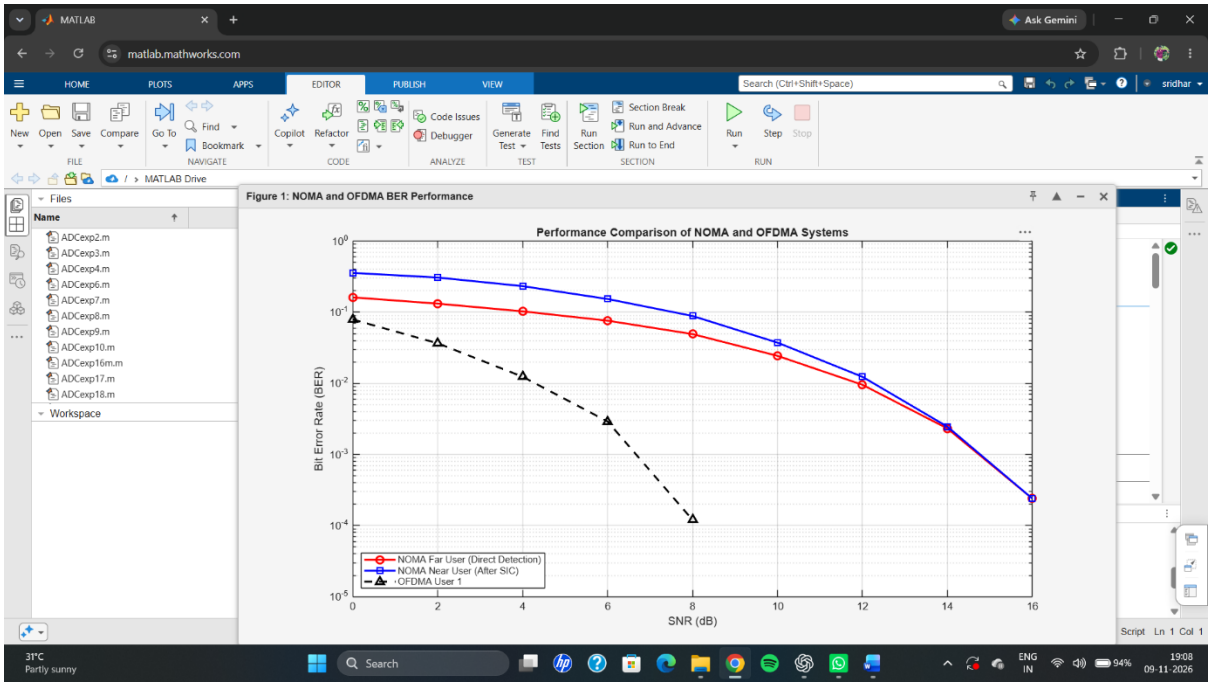
2. To implement successive interference cancellation (SIC) at the receiver for separating the superimposed user signals.

3. To simulate and compare user throughput and/or BER for selected powerallocation settings.

4. To compare the simplified NOMA performance with an equivalent OFDMA transmission case and explain the effect of orthogonal and non-orthogonal resource allocation.

PROGRAME CODE :





RESULT: The MATLAB simulation of NOMA and OFDMA systems was successfully implemented and analyzed using Successive Interference Cancellation (SIC). The BER performance of the far user, near user, and OFDMA system was evaluated over different SNR values. The results demonstrated that the bit error rate decreases as SNR increases, and SIC helps the near user eliminate inter-user interference and improve signal detection.